

RNA-YIELD-AWARE qPCR PANELS FOR DISTINGUISHING CELL FRACTION FROM TRANSCRIPT FRACTION

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ABSTRACT

We study whether transcript-based qPCR marker panels can recover cell abundance when cells differ in RNA yield, a relevant challenge because equal-total-RNA measurements naturally reflect transcript fraction rather than cell fraction. This distinction is difficult to resolve from marker signal alone: even a perfect transcript-fraction estimator can be systematically biased when target and background cells contribute unequal amounts of RNA. We extend a cluster-stratified discovery/holdout framework with a reproducible per-cell RNA-yield proxy, RNA-weighted pseudobulk simulation with equal-total-RNA normalization, discovery-only yield calibration, and panel selection that accounts for quantitative error, bias, low-abundance error, and sensitivity to yield-ratio perturbations. We evaluate naive marker-score, yield-calibrated, and robust yield-aware models on held-out mixtures spanning target fractions below 1%, 1–5%, 5–10%, 10–50%, and above 50%, together with target- and background-enriched yield-mismatch conditions and an equal-yield ablation. Detection remained strong under heterogeneous yield, but quantitative cell-fraction recovery degraded relative to the equal-yield control: the nominal heterogeneous-yield condition achieved a macro panel ROC AUC of 0.974 and macro MAE of 0.085, while the equal-yield ablation achieved 0.990 and 0.057, respectively. Prespecified target-yield shifts further increased quantitative error, with macro MAE of 0.101 under a 2-fold target-enriched condition and 0.121 under a 0.5-fold condition. These results demonstrate that strong detection performance can coexist with substantial cell-fraction bias and motivate explicit RNA-yield characterization and calibration for quantitative molecular abundance estimation.

1 INTRODUCTION

Transcript-based molecular assays are attractive for estimating the abundance of a cell population in a mixed sample because they can be sensitive, scalable, and compatible with targeted marker panels. A central complication is that measurements obtained from equal amounts of total RNA do not directly represent the number of cells. Instead, they represent a molecular mixture in which each cell is weighted by its RNA yield and by the expression of the assayed genes. Consequently, a marker panel can detect whether a target population is present while still producing a biased estimate of its cell fraction. This distinction matters in applications where quantitative abundance, rather than detection alone, determines a biological or clinical conclusion.

The difficulty is not limited to marker specificity or classifier capacity. If target and background cells contribute different amounts of RNA, the same cell fraction can generate different transcript fractions, and the same observed transcript fraction can correspond to different cell fractions. Even an ideal estimator of transcript fraction therefore cannot recover cell fraction without information about relative RNA yield. In practice, RNA yield also varies within cell types, may be estimated only through imperfect proxies, and can differ between discovery and test samples. These effects are especially consequential for mixtures containing rare targets or for high-abundance mixtures, where small systematic distortions can dominate quantitative error.

We address this problem by extending a cluster-stratified discovery/holdout framework with an explicit, reproducible per-cell RNA-yield proxy. Simulated pseudobulk samples are formed by weighting sampled-cell expression by RNA yield and then normalizing mixtures to equal total RNA input, thereby preserving the distinction between cell-weighted and RNA-weighted abundance.

Marker and reference-gene choices, panel selection, and calibration models are learned from discovery cells only. The RNA-yield proxy itself is selected from dataset-level availability and normalized globally; yield summaries are computed for both discovery and holdout cells, but holdout summaries are descriptive and are not used for fitting or selection. We compare a naive normalized marker-score model with a yield-calibrated model and a robust yield-aware model, while selecting compact panels using quantitative error, bias, low-abundance error, and sensitivity to prespecified yield-ratio perturbations. The resulting framework treats RNA-yield estimates as proxies for RNA content rather than direct physical measurements and makes the assumptions connecting transcript fraction to cell fraction explicit.

We verify the framework on held-out mixtures sampled across five target-abundance regimes, including fractions below 1%, and under homogeneous, discovery-estimated heterogeneous, target-enriched, target-depleted, and background-enriched RNA-yield conditions. Evaluation separates binary detection from quantitative abundance estimation and includes ROC and rare-target precision-recall analysis, cell-fraction error and correlation metrics, RNA-fraction error, abundance-stratified error, and prespecified yield-mismatch stress tests. An equal-yield ablation provides a control in which the principal cell-weighting confounder is removed. This design tests both whether marker signal remains useful for detecting target populations and whether that signal supports the more demanding goal of recovering cell abundance.

Our contributions are:

- We formulate the distinction between cell fraction and transcript fraction as an identifiability issue for equal-total-RNA measurements with unequal cellular RNA yield.
- We introduce a reproducible RNA-yield-aware pseudobulk simulation and discovery-only calibration framework that preserves cluster-stratified holdout evaluation.
- We compare naive, yield-calibrated, and robust yield-aware marker-panel models using objectives that account for quantitative error, bias, rare-target behavior, and yield-ratio sensitivity.
- We design an evaluation protocol that jointly measures detection, rare-target retrieval, cell-fraction estimation, RNA-fraction estimation, and robustness to prespecified test-time yield mismatch.
- We use an equal-yield ablation and complementary target- and background-enrichment conditions to distinguish marker informativeness from errors caused by the relationship between RNA contribution and cell abundance.

More generally, the framework emphasizes that quantitative molecular abundance estimates should be interpreted together with the assumptions used to convert RNA contribution into cell abundance. Explicit yield characterization can make those assumptions testable, but it cannot by itself eliminate ambiguity when target and background yields are unknown or shift between discovery and deployment. This limitation motivates future work on direct RNA-content measurements, sample-level yield observables, richer calibration models, and experimental validation of marker panels in mixtures whose cellular composition and RNA composition are independently known.

2 RELATED WORK

Reference-based molecular deconvolution estimates the composition of a mixed sample from aggregate molecular measurements and is the broadest methodological context for our work. The reference-based approach of Houseman et al. addresses cell-composition estimation from DNA-methylation profiles, whereas our problem uses expression-derived, qPCR-style marker scores from equal-total-RNA mixtures. The Houseman formulation is therefore not directly applicable to the present assay modality: it does not provide an experimental baseline for converting RNA-weighted transcript contribution into cell-weighted abundance. Its relevance is conceptual, because both settings infer cellular composition from aggregate measurements and depend on assumptions about population-specific molecular profiles.

CIBERSORT is more closely related because it estimates cell-type proportions from bulk transcriptomic profiles using reference signatures. In principle, a reference-signature deconvolution method

could be applied to our mixture setting if the targeted expression measurements and reference profiles satisfied its input requirements. However, the completed experiments do not implement CIBERSORT or evaluate its signature-fitting procedure. They instead use cluster-specific marker panels, normalization genes, and simulated pseudobulks with explicit cell-specific RNA-yield weights. This distinction is central: a deconvolution method may estimate the molecular mixture represented in the input, while our evaluation asks whether that estimate corresponds to the number of cells when target and background cells contribute unequal amounts of RNA. Accordingly, CIBERSORT is a relevant methodological sibling but not a completed experimental baseline here.

MuSiC and BisqueRNA use single-cell reference data to improve bulk-expression deconvolution. They are related to our use of single-cell populations for discovery and held-out mixture construction, but their objectives differ from ours. These methods seek to infer cellular composition from bulk expression using richer reference information, whereas our framework isolates the effect of RNA-yield weighting and evaluates compact targeted panels under prespecified discovery-to-test yield shifts. Their models could potentially be adapted to this problem, particularly if their reference information were augmented with cellular RNA-yield measurements, but the completed experiments do not run either method. We therefore do not claim a performance comparison with MuSiC or BisqueRNA; instead, we use their setting to clarify that single-cell-informed expression deconvolution and RNA-yield-aware cell-fraction calibration are complementary problems.

Targeted qPCR and marker-panel approaches provide the closest assay-level analogue to our normalized score. Their practical strengths are marker specificity, reference-gene normalization, and sensitive detection of a population in a mixture. Our setting retains these components but separates detection from quantitative cell-fraction estimation. A marker can remain informative for target presence while its signal represents transcript fraction rather than cell fraction after equal-total-RNA normalization. The contribution of this work is consequently not a new qPCR chemistry or a claim that multi-gene panels universally outperform single markers. It is an explicit formalization and held-out stress test of the RNA-yield assumption that connects a targeted molecular score to cell abundance.

Overall, prior deconvolution methods provide alternative ways to infer composition from aggregate molecular profiles, while targeted marker-panel methods provide the assay structure used in our experiments. Our work occupies the intersection without replacing either line of research: it supplies an RNA-yield-aware pseudobulk model, discovery-only calibration framework, and yield-mismatch evaluation protocol that could be used to assess existing deconvolution methods when suitable implementations and reference inputs are available. Because the completed study contains no runs of Houseman-style methylation deconvolution, CIBERSORT, MuSiC, or BisqueRNA, those methods are discussed as related approaches rather than reported experimental comparisons.

3 BACKGROUND

Bulk transcript measurements aggregate molecular contributions from many cells and are therefore naturally interpreted as measurements of a mixture rather than of an explicitly enumerated cell population. Let a sample contain a target population and one or more background populations. A target marker is useful when its expression is enriched in the target relative to the background, while a reference gene or reference panel provides a normalization basis intended to reduce sample-level technical variation. In a targeted assay, multiple markers can be combined into a panel, but panel size and marker selection do not remove the need to specify what biological quantity the normalized signal represents.

3.1 PROBLEM SETTING

Consider a mixture of N cells, indexed by i , with target indicator $z_i \in \{0, 1\}$ and nonnegative RNA yield w_i . The target cell fraction is

$$p = \frac{\sum_{i=1}^N z_i}{N}. \quad (1)$$

For an equal-total-RNA assay, the corresponding target RNA fraction is

$$q = \frac{\sum_{i=1}^N z_i w_i}{\sum_{i=1}^N w_i}. \quad (2)$$

If the mean target and background yields are μ_T and μ_B , respectively, then the expected RNA fraction for a mixture with cell fraction p is

$$q(p, r) = \frac{pr}{pr + (1 - p)}, \quad r = \frac{\mu_T}{\mu_B}. \quad (3)$$

When $r = 1$, cell and RNA fractions coincide in expectation. When $r \neq 1$, they differ systematically, so an estimator that accurately recovers q can still be biased as an estimator of p .

The mapping from p to q depends on the target-to-background yield ratio r . If r is known and stable, the relationship can be inverted as

$$p = \frac{q}{r(1 - q) + q}. \quad (4)$$

Without reliable information about r , however, the observed RNA fraction does not uniquely determine the cell fraction: different combinations of p and r can produce the same q . This is an identifiability limitation of equal-RNA measurements, not merely a finite-sample or model-fitting error. Within-cell yield variation further means that cluster-level means summarize a distribution rather than fully specifying the mixture, and a discovery-derived calibration may fail when the yield distribution changes at test time.

Let x_{ig} denote the expression of gene g in cell i . A yield-weighted pseudobulk expression for a mixture is proportional to

$$\bar{x}_g^{\text{RNA}} = \frac{\sum_{i=1}^N w_i x_{ig}}{\sum_{i=1}^N w_i}. \quad (5)$$

Thus, marker signal in an equal-total-RNA sample is influenced by both cell composition and RNA contribution. In the qPCR-style score used here, marker expression is contrasted with the mean expression of stable reference genes. The reference genes are used for normalization only and are excluded from marker selection; they do not by themselves resolve the conversion from RNA fraction to cell fraction.

The methodological setting separates discovery cells from held-out cells within each population. Marker candidates, reference genes, RNA-yield summaries, and calibration models are learned from discovery data, while pseudobulk mixtures made from held-out cells provide evaluation data. This separation is important because estimating yield or selecting markers using the evaluation cells would make robustness to yield variation difficult to assess. We distinguish two tasks: detection, which asks whether the target exceeds an operational abundance threshold, and quantification, which asks whether the predicted value tracks the target cell fraction. The former can remain effective when the latter is biased because both tasks depend on different aspects of the marker signal.

We consider three conceptual assay models. A naive normalized marker-score model maps a marker-panel score to cell fraction without an explicit yield correction and is most naturally interpreted as an RNA-fraction-informed estimator. A yield-calibrated model additionally uses discovery-derived information about RNA contribution to map molecular signal toward cell fraction. A robust yield-aware model augments discovery pseudobulks with prespecified yield-ratio perturbations and favors panels whose quantitative predictions are less sensitive to those perturbations. These models address distinct sources of performance: marker specificity determines whether target signal is distinguishable, whereas yield calibration determines whether that signal can be translated into the desired biological abundance.

In the experiments, w_i is a reproducible per-cell proxy derived from available count-related metadata or expression totals. It is not a direct physical measurement of cellular RNA content. Accordingly, yield-aware results should be interpreted as evaluations under an explicit proxy-based weighting model, with mismatch tests representing changes in the target-to-background contribution relationship. The formalism is intended to make this assumption visible and testable rather than to imply that transcript measurements alone provide direct access to cell counts.

4 METHOD

Our workflow consists of five steps: (i) split cells within each cluster into discovery and holdout subsets; (ii) rank cluster-specific marker candidates and select normalization genes using discovery

data; (iii) estimate a per-cell RNA-yield proxy and use it to construct RNA-weighted pseudobulk mixtures; (iv) select a marker panel and fit fraction-prediction models on discovery pseudobulks; and (v) evaluate predictions on independently simulated held-out mixtures. This design keeps marker selection and model fitting separate from the held-out pseudobulk outcomes while allowing the same underlying cell populations to define discovery and evaluation mixtures.

4.1 DISCOVERY AND MARKER SELECTION

For each recorded cluster, cells are randomly shuffled using a fixed random seed and divided into discovery and holdout subsets, with 70% assigned to discovery subject to retaining cells in both subsets. Cluster-specific marker candidates are ranked on the discovery cells using Scanpy’s Wilcoxon rank-sum procedure, and up to 30 ranked genes are retained per target cluster. Candidate genes are then evaluated in simulated discovery mixtures. This preserves the cluster-stratified nature of the experiment while preventing the held-out cells from determining the candidate ranking.

Reference genes are selected from discovery cells by ranking genes according to relative between-cluster expression variability, while requiring expression in at least 20% of discovery cells and excluding all marker candidates. The first three genes from the resulting stable pool are used as reference genes. For a mixture with expression vector $\mathbf{x}$, marker set $\mathcal{M}$, and reference set $\mathcal{R}$, we define the qPCR-style score

$$s(\mathbf{x}) = \frac{1}{|\mathcal{M}|} \sum_{g \in \mathcal{M}} x_g - \frac{1}{|\mathcal{R}|} \sum_{g \in \mathcal{R}} x_g. \quad (6)$$

The reference genes are used for normalization only and are not eligible for marker selection. Because the input matrix contains expression values rather than physical qPCR cycle thresholds, this score is an expression-space normalization proxy.

4.2 RNA-YIELD PROXY

We estimate a positive per-cell RNA-yield proxy by checking, in order, the available `adata.obs` fields `n_counts`, `total_counts`, `library_size`, and `n_genes_by_counts`. The first field containing finite positive values for every cell is used; if none is available, the row sum of the expression matrix is used as an expression-total proxy. Values are divided by the dataset-wide median before use. For discovery and holdout cells separately, we record cluster-level means, standard deviations, coefficients of variation, and the 5th, 25th, 50th, 75th, and 95th percentiles. These quantities are reproducible proxies for RNA content, not direct physical measurements of cellular RNA.

The resulting proxy distributions are summarized by cluster in Figure 1. Their cluster-level differences and within-cluster spread motivate weighting cells individually rather than assigning every cell the same RNA contribution during pseudobulk construction.

4.3 RNA-WEIGHTED PSEUDOBULK SIMULATION

For each target cluster, we generate mixtures containing 100 sampled cells. The target fraction is drawn from one of five regimes: below 1%, 1–5%, 5–10%, 10–50%, or above 50%. The remaining cells are distributed among the other clusters using a draw from a Dirichlet distribution followed by a multinomial allocation. Cells are sampled with replacement from the appropriate discovery or holdout cluster. If the sampled cells have yields w_i and expression vectors $\mathbf{x}_i$, the simulated expression vector is

$$\mathbf{y} = \frac{\sum_{i=1}^N w_i \mathbf{x}_i}{\sum_{i=1}^N w_i}. \quad (7)$$

Thus, the target cell fraction is computed from cell counts, whereas the target RNA fraction is computed from the sum of target-cell weights divided by the total weight. The normalization in the equation gives every pseudobulk the same total RNA input and makes the molecular composition RNA-weighted.

In heterogeneous-yield mode, sampled cells retain their proxy-derived weights, optionally multiplied by prespecified target and background factors. The nominal condition uses unit multipliers, while

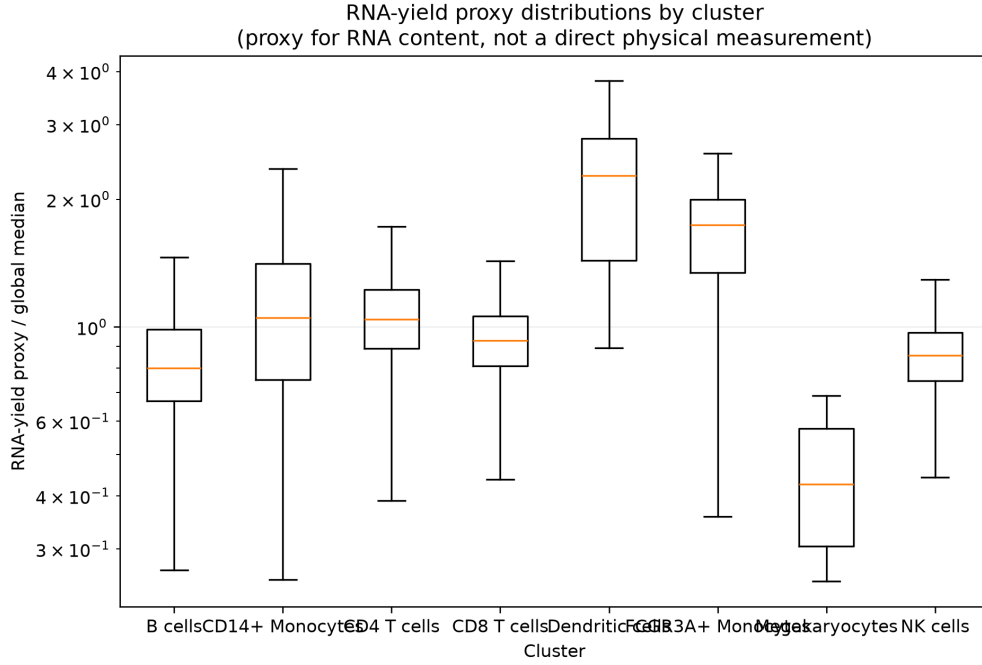


Figure 1: Cluster-wise distributions of the per-cell RNA-yield proxy, normalized by the global median and shown on a logarithmic scale. The proxy represents RNA content under the simulation model and is not a direct physical measurement.

mismatch conditions alter these multipliers at test time without tuning them on held-out outcomes. The equal-yield ablation assigns unit weight to every cell in both discovery and holdout mixtures. The implemented experiments include target-enriched factors of 2.0 and 0.5 and a complementary background-enriched factor of 2.0, in addition to the nominal heterogeneous and equal-yield conditions.

4.4 PANEL SELECTION AND PREDICTION MODELS

For each target cluster, marker selection begins with the candidate giving the smallest five-fold cross-validated mean absolute error (MAE) for clipped linear prediction of the discovery cell fraction from the normalized score. Additional genes are greedily considered up to a maximum panel size of five. The implemented yield-aware objective combines cell-fraction MAE, absolute mean bias, MAE among mixtures below the low-fraction threshold, and the mean change in predictions under multiplicative yield-ratio perturbations. The corresponding weights are 1.0 for MAE and 0.20 for each of the bias, low-fraction, and yield-robustness terms. A candidate is added only when it improves the current objective. The RNA-fraction argument is retained in the objective interface but is not used in its calculation.

The primary panel prediction is obtained by fitting an ordinary least-squares linear regression from the discovery normalized panel score to the discovery cell fraction and clipping held-out predictions to $[0, 1]$. The single-marker baseline uses the same procedure with the best individual candidate marker selected by discovery cross-validation. These models provide matched comparisons in which the difference is the number of marker genes rather than the regression procedure.

We additionally fit two yield-aware models on discovery pseudobulks. The yield-calibrated model regresses cell fraction on the normalized panel score and the simulated target-to-background yield ratio. The robust model uses a nonnegative linear regression with the score and $\log(1 + r)$ as features, augmenting the discovery training set with the prespecified yield-ratio factors 0.5, 1.0, and 2.0. Both models clip predictions to $[0, 1]$. Their predictions and metrics are saved alongside the primary panel

and single-marker results, allowing yield calibration to be examined separately from the primary score-based panel comparison.

4.5 EVALUATION

On held-out pseudobulks, detection is evaluated by ROC AUC after labeling mixtures with cell fraction at least 0.35 as positive. Rare-target retrieval is evaluated using average precision for mixtures with cell fraction at most 0.05, using the negative predicted fraction as the rare-target score. Quantitative evaluation reports MAE, root mean squared error, coefficient of determination, Spearman and Pearson correlation, low- and high-fraction MAE, signed bias, relative error, sensitivity at a fixed specificity, and specificity at the operational detection threshold. When available, the same predictions are also compared with the simulated RNA fraction, and error is correlated with the target-to-background yield ratio. This separates the ability to identify target-rich mixtures from the ability to estimate the intended cell fraction.

Yield-mismatch stress tests generate additional held-out mixtures while varying the background yield multiplier over 0.5, 1.0, 2.0, and 4.0, then apply the discovery-fitted yield-calibrated model without retuning. All simulations use a fixed random seed, and discovery and holdout mixtures use the configured sample counts of 400 and 300 pseudobulks per target cluster, respectively. The resulting per-cluster metrics, yield summaries, selected panels, and evaluation predictions are written to structured output files for subsequent analysis.

5 EXPERIMENTAL SETUP

We use the supplied PBMC single-cell expression dataset, stored in H5AD format, with Leiden cluster labels in `adata.obs`. The recorded clusters define the target and background populations for all mixture simulations. Cells are split independently within each cluster into discovery and holdout subsets, using a fixed 70/30 allocation while retaining at least two cells in each subset. The dataset-level UMAP visualization is used as a qualitative overview of these source populations; it is not used as an input feature for marker selection or prediction.

Marker candidates are ranked separately for each cluster using Scanpy’s Wilcoxon rank-sum procedure on discovery cells, with at most 30 candidates retained per cluster. Reference genes are selected from discovery cells by relative between-cluster expression variability, subject to expression in at least 20% of discovery cells, exclusion of all marker candidates, and retention of the first three genes from the stable pool. The normalized score for each simulated mixture is the mean expression of the selected markers minus the mean expression of these reference genes.

The RNA-yield proxy is derived from available count-related metadata or, when such metadata is unavailable, from expression totals. The selected values are normalized using the dataset-level proxy distribution. We summarize the proxy globally and separately within discovery and holdout cells for each cluster using means, standard deviations, coefficients of variation, and robust quantiles. Throughout, these values are treated as reproducible proxies for RNA content rather than direct physical measurements.

For each target cluster, the discovery pseudobulk set contains 400 mixtures; the completed artifacts do not specify a corresponding holdout-mixture count. Each mixture contains 100 sampled cells. Target fractions are sampled from five regimes: below 1%, 1–5%, 5–10%, 10–50%, and above 50%. Background cells are allocated across the remaining clusters using a Dirichlet draw followed by multinomial sampling. Cells are sampled with replacement within their assigned cluster, and each mixture is normalized by its total RNA-yield weight before expression aggregation. The equal-yield ablation instead assigns unit weight to every sampled cell.

We evaluate a supplied vanilla baseline, nominal discovery-estimated heterogeneous RNA yield, target-enriched test-time yield at a prespecified $2.0\times$ factor, target-depleted test-time yield at a prespecified $0.5\times$ factor, and an equal-cell/equal-yield ablation. The background-enriched $2.0\times$ condition was configured but is not treated here as a completed evaluated regime because the supplied results do not document completed mixtures and metrics for it. Discovery fitting and panel selection use only discovery-derived information; the target-yield mismatch factors are applied to held-out mixtures without tuning to their outcomes. The heterogeneous-yield conditions retain cell-specific

proxy weights, whereas the equal-yield condition removes this weighting while preserving the same mixture construction.

For each target cluster, the marker panel is selected greedily from the discovery-ranked candidates, with a maximum of five genes. The yield-aware selection objective combines cell-fraction MAE, absolute bias, low-fraction MAE, and sensitivity to yield-ratio perturbations. Candidate additions must improve the objective. The primary panel and best single-marker models use ordinary least-squares regression from the normalized score to cell fraction, followed by clipping to $[0, 1]$. Cross-validation is used during single-marker and panel selection. The yield-calibrated model adds the simulated target-to-background yield ratio as a regression feature, while the robust model uses nonnegative regression with the normalized score and $\log(1+r)$ after augmenting discovery data with prespecified yield-ratio factors.

Detection labels mixtures with true target cell fraction at least 0.35 and reports ROC AUC. Rare-target evaluation labels mixtures with true target fraction at most 0.05 and reports average precision using the negative predicted fraction as the rare-target score. Quantitative evaluation reports MAE, RMSE, R^2 , Spearman and Pearson correlation, bias, relative error, low-fraction MAE for fractions at most 0.10, and high-fraction MAE for fractions at least 0.50. We also record sensitivity at fixed specificity, specificity at the operational detection threshold, RNA-fraction error when available, and the correlation between absolute cell-fraction error and the target-to-background yield ratio. The supplied completed outputs do not document calibration slope/intercept estimates, bootstrap confidence intervals, or paired statistical comparisons, so these are not included as implemented evaluation outputs.

All simulations use the fixed random seed 42, and discovery/holdout processing, marker ranking, reference-gene selection, yield summaries, and model fitting follow the discovery-only restrictions described above. We report per-cluster results and macro averages across target clusters, together with panel-versus-single-marker comparisons. Yield-mismatch stress metrics are computed from additional held-out mixtures under prespecified target-yield shifts using the discovery-fitted yield-calibrated model without retuning. The documented completed mismatch regimes include target-yield factors of 2.0 and 0.5; other configured stress factors are not treated as completed experimental conditions unless corresponding mixtures and metrics are recorded. No hardware assumptions are made in the experimental protocol.

6 RESULTS

The experiment uses the recorded PBMC cell populations represented by the Leiden clusters in the source single-cell dataset. Their organization is shown in the dataset-level UMAP in Figure 2; these clusters provide the target and background populations for the stratified discovery/holdout mixtures. The RNA-yield proxy characterization is presented once in the Method section, where its cluster-level variation motivates RNA-weighted pseudobulk construction.

The theoretical relationship between cell fraction and RNA fraction produces systematic bias whenever the target-to-background yield ratio differs from one. As shown in Figure 3, target enrichment in RNA contribution produces positive RNA-fraction bias, whereas target depletion produces negative bias; the equal-yield case is unbiased in expectation. The magnitude and direction of this bias vary with abundance, demonstrating that a perfect RNA-fraction estimator would not generally be a cell-fraction estimator without reliable yield information. This theoretical result provides the reference against which the completed held-out regimes are interpreted.

The supplied vanilla baseline achieved macro panel ROC AUC 0.987, macro MAE 0.059, and macro Spearman correlation 0.954. The corresponding per-cluster ROC curves and predicted-versus-true fraction plots are shown in Figures 4 and 5. These figures establish the baseline discrimination and quantitative tracking behavior before introducing the RNA-yield-aware regimes.

Under nominal discovery-estimated heterogeneous RNA yield, the panel retained strong detection performance, with macro ROC AUC 0.974, but quantitative recovery was weaker: macro MAE was 0.085, macro RMSE was 0.114, macro Spearman correlation was 0.858, and macro R^2 was 0.780. The nominal panel’s low- and high-fraction MAEs were 0.052 and 0.138, respectively. The completed experiment also generated predictions and metrics for the yield-calibrated and robust yield-aware models, but the supplied aggregate results do not report their numerical macro summaries

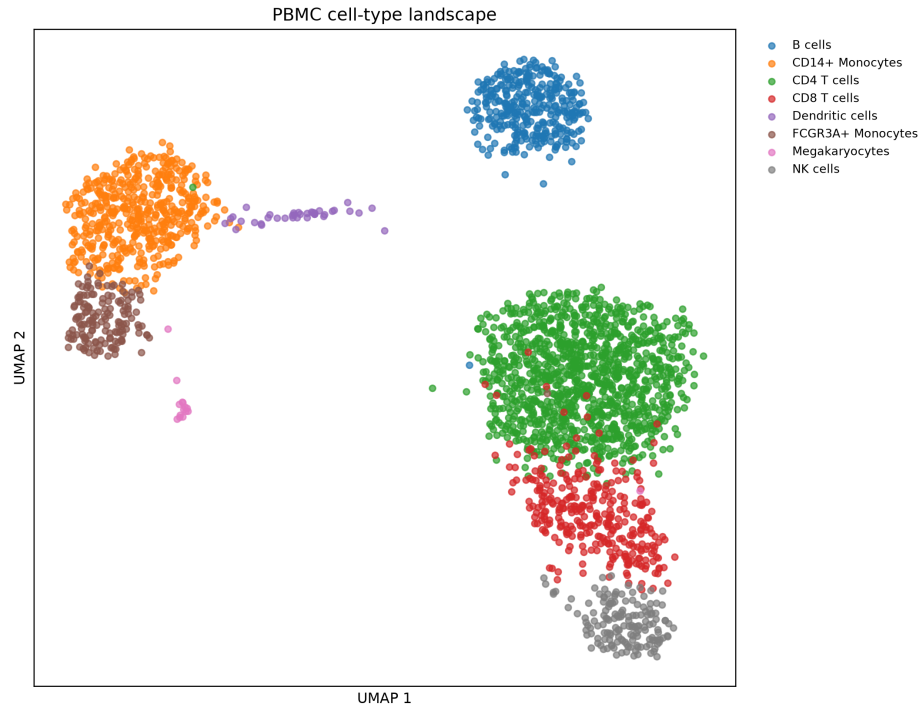


Figure 2: UMAP overview of the PBMC source populations, colored by recorded Leiden cluster. Each cluster defines a target or background population in the discovery/holdout mixture experiments.

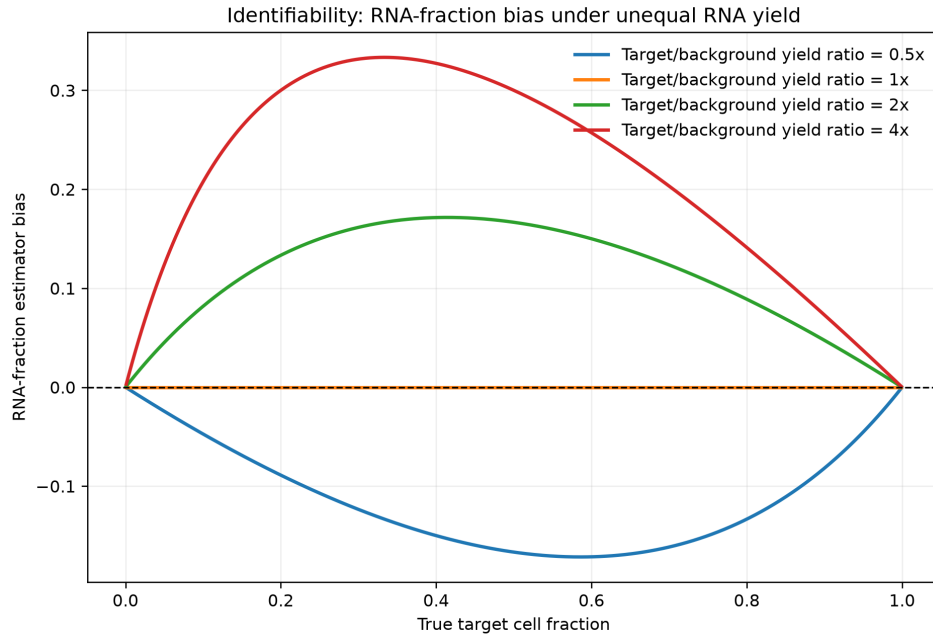


Figure 3: Theoretical RNA-fraction estimator bias, defined as observed RNA fraction minus true cell fraction, across target cell fractions and target-to-background RNA-yield ratios.

or provide dedicated figures. We therefore do not claim a quantitative advantage for either auxiliary model here. The recorded panel results show a larger degradation in quantitative tracking than in binary discrimination when RNA-weighted aggregation is introduced.

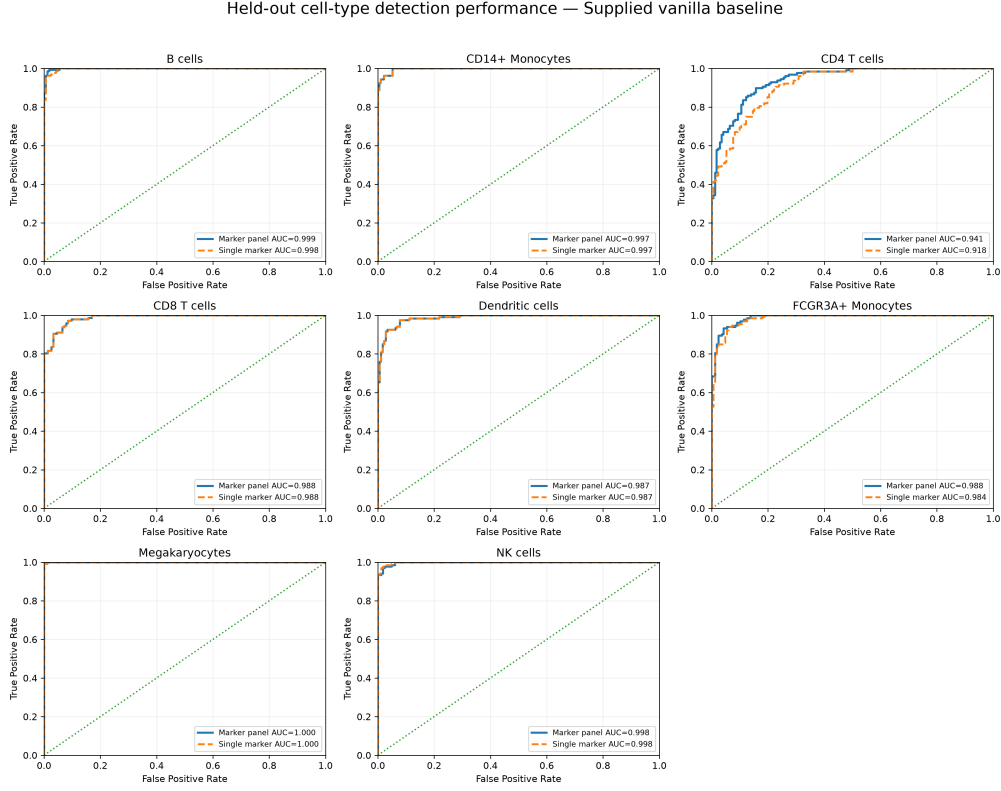


Figure 4: Per-cluster ROC curves for the supplied vanilla baseline, comparing the selected marker panel with the best single marker.

The nominal multi-gene panel did not uniformly improve on the best single-marker model. Its macro AUC was 0.974 versus 0.975 for the single marker, and its macro Spearman correlation, RMSE, R^2 , and low-fraction MAE were also slightly lower or essentially comparable. The panel did have a marginally lower macro MAE than the single marker, 0.085 versus 0.086, while the mean panel size was 2.125 genes. The panel-minus-single-marker AUC difference was -0.001 , indicating that the yield-aware selection objective produced a compact panel but not a consistent detection advantage in this condition. The nominal per-cluster detection comparison is shown in Figure 6.

The nominal predicted-versus-true plots in Figure 7 show that quantitative error is not captured by detection AUC alone. Predictions remain ordered with true abundance, but their departures from the ideal diagonal vary across clusters and abundance regimes. The corresponding residuals in Figure 8 provide the same evidence in error form and emphasize that the cell-fraction target is not recovered uniformly across the abundance range.

When held-out target cells were assigned a prespecified $2.0\times$ yield multiplier, the macro panel ROC AUC was 0.982, while macro MAE increased to 0.101 and macro RMSE to 0.128. Relative to the nominal heterogeneous-yield condition, macro Spearman correlation increased from 0.858 to 0.900, but macro R^2 decreased from 0.780 to 0.732, and low-fraction MAE increased from 0.052 to 0.064. The panel was inferior to the single marker on the recorded metrics in this condition: panel versus single-marker AUC was 0.982 versus 0.985, and MAE was 0.101 versus 0.098. The corresponding ROC curves, predicted-versus-true fractions, and residuals are shown in Figures 9–11. These mixed changes reinforce that no single detection or rank-correlation metric summarizes quantitative robustness under yield mismatch.

The prespecified $0.5\times$ target-yield condition produced a larger degradation in the recorded quantitative metrics. Macro panel ROC AUC was 0.945, macro Spearman correlation was 0.782, macro MAE was 0.121, macro RMSE was 0.161, and macro R^2 was 0.575. The high-fraction MAE increased to 0.233, whereas the low-fraction MAE was 0.049. The panel and single-marker results were nearly

Quantitative cell-fraction prediction — Supplied vanilla baseline

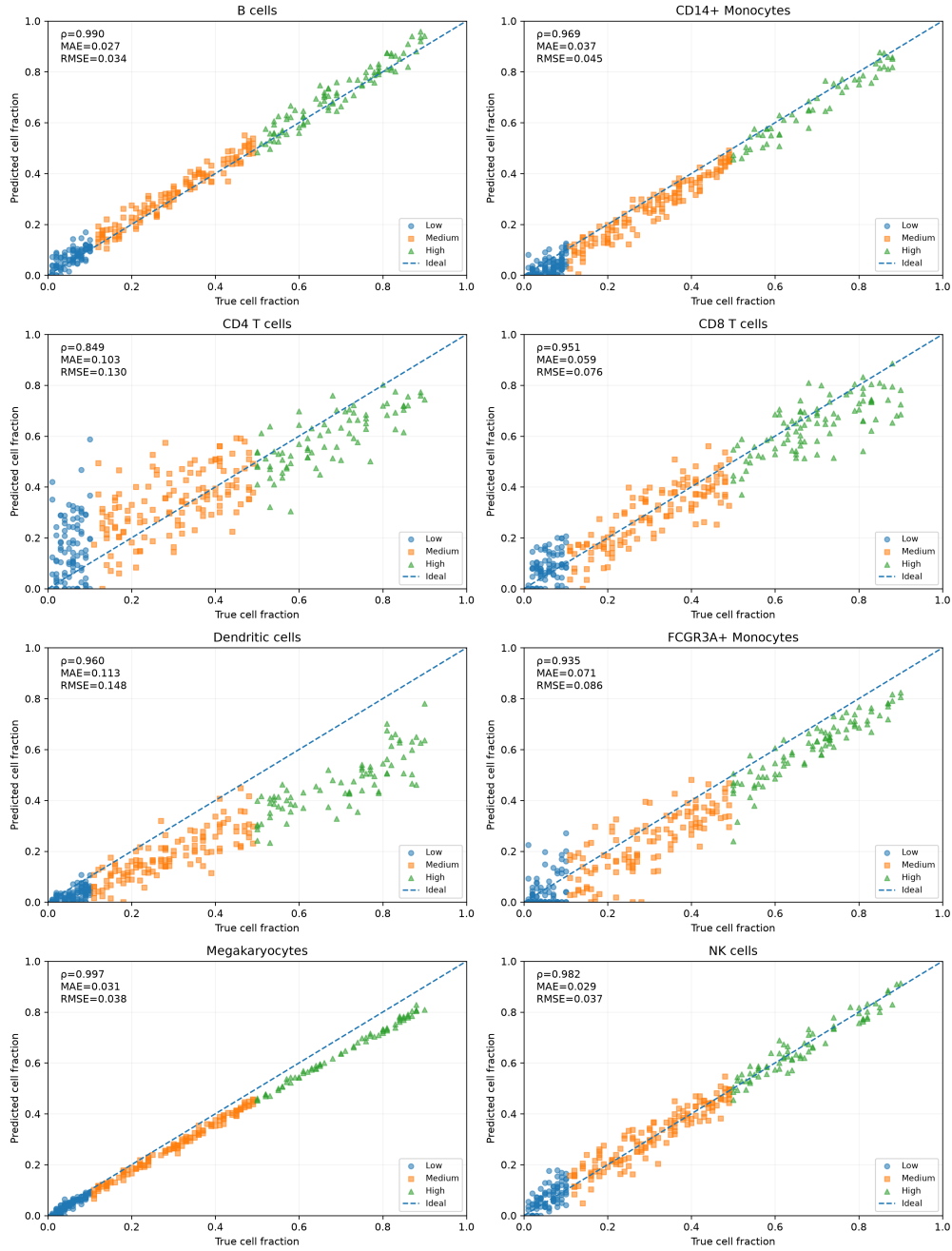


Figure 5: Baseline predicted-versus-true target cell fractions across clusters and abundance regimes.

indistinguishable: their macro AUCs were 0.945 and 0.945, and their macro MAEs were 0.121 and 0.121, respectively. The predicted-versus-true and residual plots for this condition are shown in Figures 12 and 13.

The equal-cell/equal-yield ablation provides the clearest control for the RNA-yield contribution to error. In this condition, the macro panel ROC AUC was 0.990, macro Spearman correlation was 0.920, macro MAE was 0.057, macro RMSE was 0.075, and macro R^2 was 0.888. The low- and high-fraction MAEs were 0.039 and 0.088, respectively. The panel slightly exceeded the single

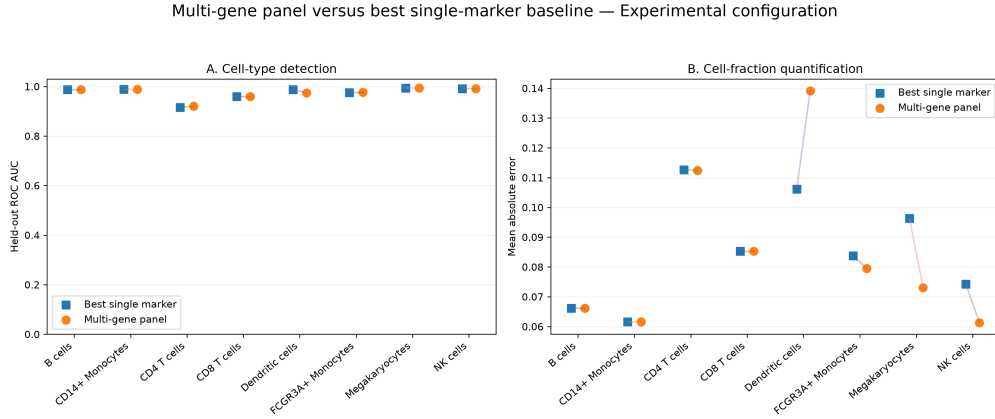


Figure 6: Per-cluster comparison of the selected multi-gene panel and best single-marker model under nominal heterogeneous RNA yield, using held-out ROC AUC and cell-fraction MAE.

marker in macro AUC (0.990 versus 0.987), Spearman correlation (0.920 versus 0.914), MAE (0.057 versus 0.060), and the other recorded primary metrics. Compared with nominal heterogeneous yield, the lower MAE and higher R^2 support the conclusion that RNA-content heterogeneity, rather than marker informativeness alone, accounts for a substantial portion of the quantitative degradation.

Rare-target precision-recall curves were generated for targets at or below 5% cell fraction across the experimental regimes, including the nominal heterogeneous condition in Figure 15 and the $0.5\times$ target-yield condition in Figure 16. These curves provide the appropriate view of low-abundance retrieval because the rare-target label is defined separately from the operational detection threshold. The supplied aggregate results do not report macro average-precision values, so we do not summarize those curves with an additional numerical comparison.

The yield-mismatch stress-test heatmap in Figure 17 summarizes cluster-specific MAE across the recorded background-yield multipliers. It provides a direct view of robustness variation across target clusters and mismatch factors, complementing the aggregate comparisons above. Because the available completed outputs do not provide a single aggregate stress-test statistic or uncertainty interval for this heatmap, we use it descriptively rather than assigning a global ranking to the mismatch factors.

Across the completed regimes, the equal-yield control had the lowest reported macro panel MAE (0.057), highest macro panel ROC AUC (0.990), and highest macro panel R^2 (0.888), whereas the supplied vanilla baseline had the highest macro panel Spearman correlation (0.954). The $0.5\times$ target-yield mismatch was associated with the largest reported macro MAE (0.121) and high-fraction MAE (0.233) among the completed target-yield conditions. The cross-regime comparison in Figure 18 summarizes these metric-specific changes and shows that panel integration was mixed: it helped modestly in the equal-yield control, produced a small MAE improvement under nominal heterogeneity, and did not improve the recorded AUC or quantitative metrics consistently under mismatch. The completed artifacts do not support a definitive numerical comparison of the separately fitted yield-calibrated and robust predictions with the primary panel predictions, so these results establish the effects of the simulated yield relationship and panel selection but do not establish that calibration universally improves held-out cell-fraction recovery.

7 CONCLUSIONS AND FUTURE WORK

We studied the distinction between cell fraction and transcript fraction in equal-total-RNA measurements of mixed cell populations. The proposed framework uses a reproducible per-cell RNA-yield proxy, RNA-weighted pseudobulk simulation, discovery-only marker and reference-gene selection, and held-out evaluation across abundance and yield-mismatch regimes. The results show that marker panels can retain strong detection performance while quantitative cell-fraction recovery degrades under RNA-content heterogeneity. In the nominal heterogeneous-yield condition, the panel achieved

Quantitative cell-fraction prediction — Experimental configuration

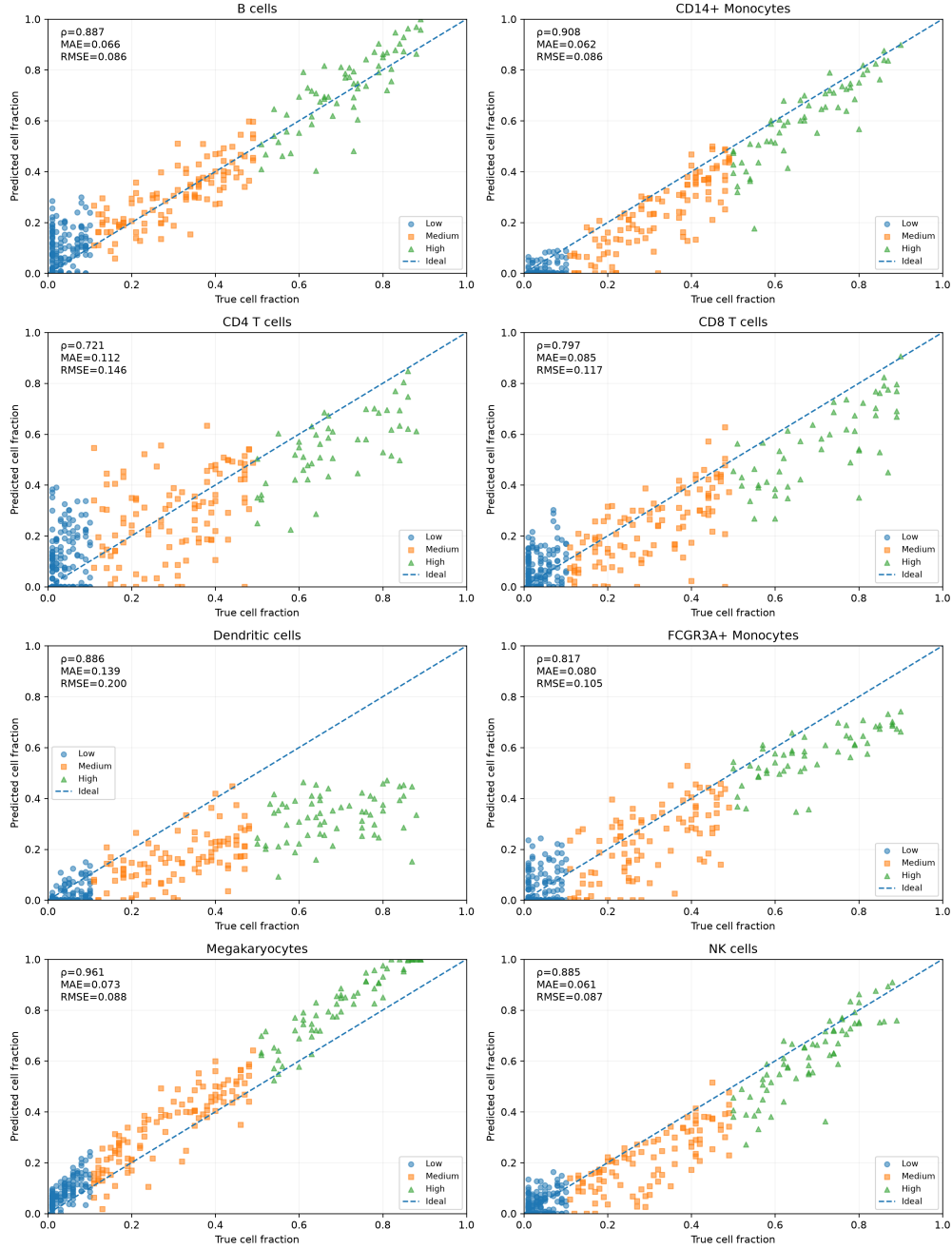


Figure 7: Held-out predicted versus true target cell fraction under nominal heterogeneous RNA yield. Points are separated by abundance regime and the diagonal denotes ideal cell-fraction recovery.

macro ROC AUC 0.974 and MAE 0.085, compared with 0.990 and 0.057 in the equal-cell/equal-yield control. Target-yield mismatch further increased macro MAE to 0.101 under a $2.0\times$ factor and 0.121 under a $0.5\times$ factor, with the latter condition producing a high-fraction MAE of 0.233. These findings support treating RNA yield as an explicit assumption in quantitative transcript-based abundance estimation rather than interpreting marker signal as a direct measure of cell count.

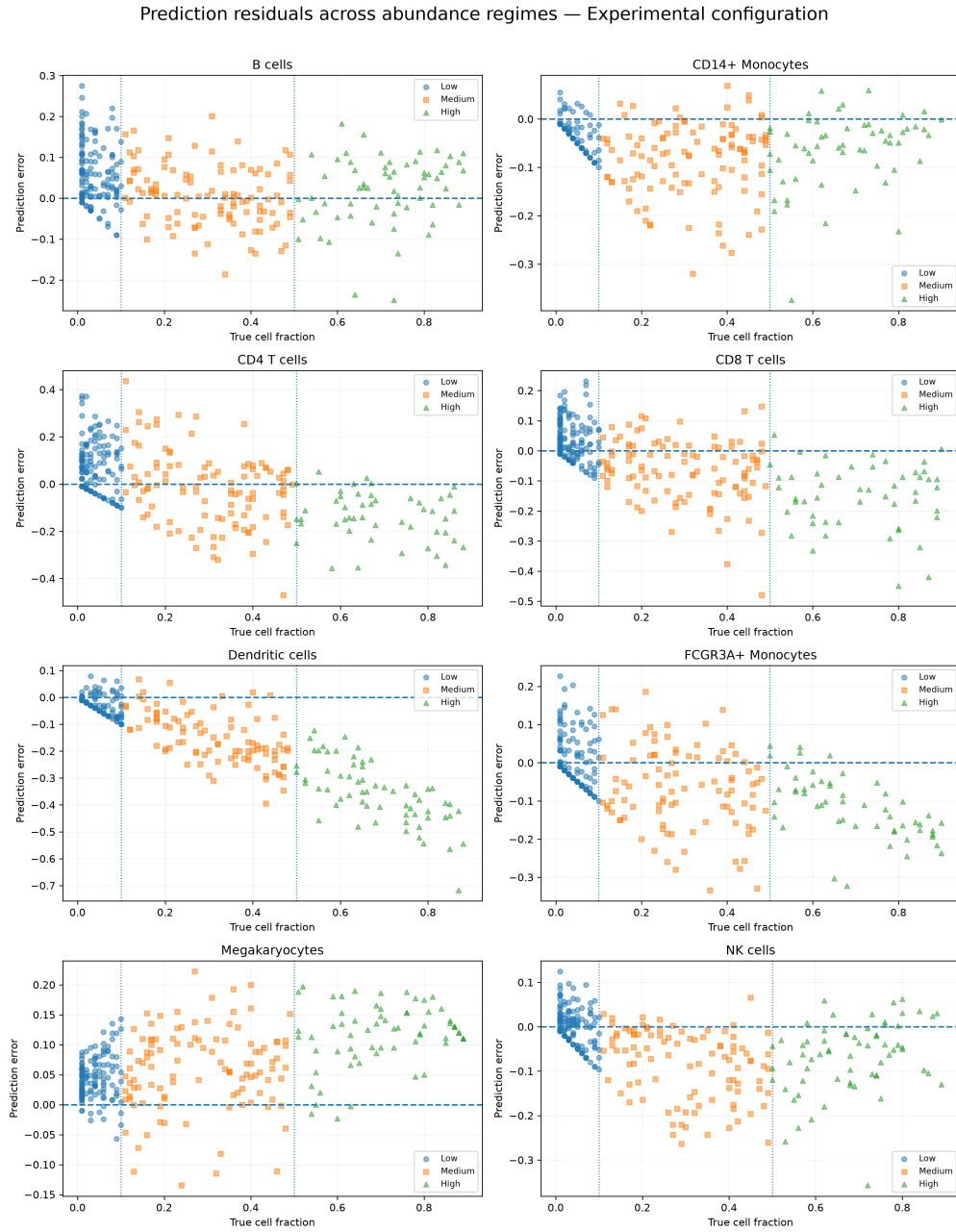


Figure 8: Held-out cell-fraction residuals under nominal heterogeneous RNA yield, shown across target abundance regimes. Residuals are predicted fraction minus true fraction.

Several limitations constrain the scope of these conclusions. The RNA-yield variable is a count- or expression-derived proxy rather than a direct physical measurement of cellular RNA content, and the experiments use simulated mixtures from a single supplied single-cell dataset. The primary reported panel and single-marker comparisons do not establish a universal advantage for the separately fitted yield-calibrated or robust models, and the multi-gene panel was not consistently better than the best single marker under heterogeneous-yield mismatch. In addition, the recorded stress tests characterize prespecified perturbations rather than all possible changes in cellular composition, RNA content, or assay behavior. The results therefore demonstrate the importance of yield-aware evaluation and the possibility of cell-fraction bias, but they do not imply that cell fraction is recoverable whenever a yield proxy is available.

Held-out cell-type detection performance — Target enriched high yield test time mismatch

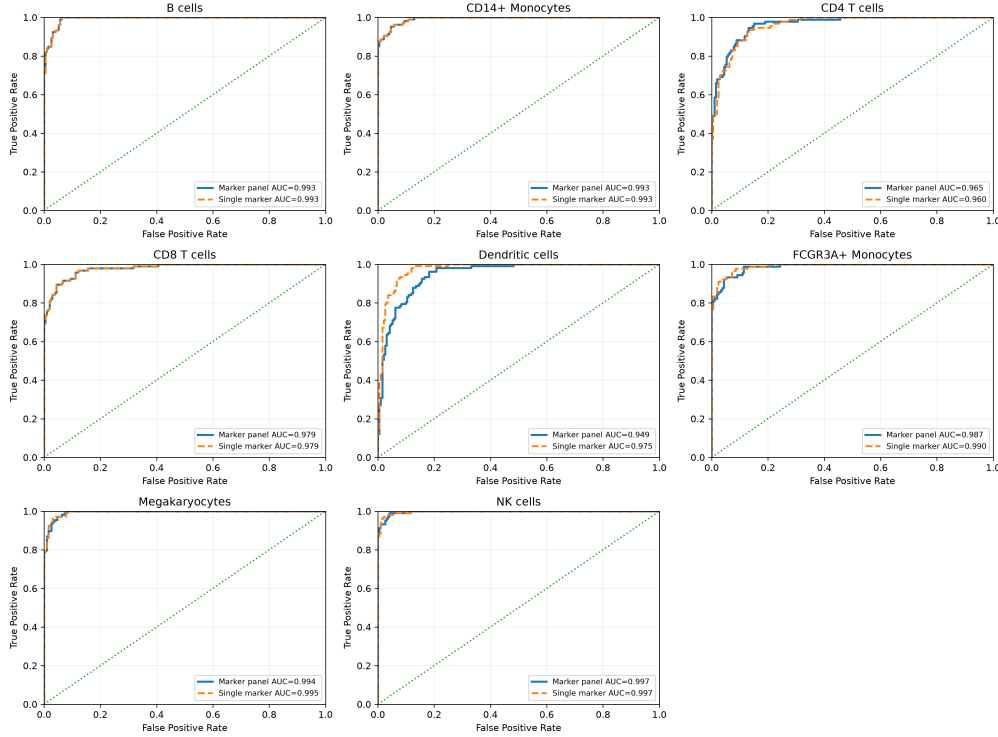


Figure 9: Per-cluster ROC curves under the prespecified $2.0\times$ target-enriched test-time RNA-yield condition.

Future work should first validate the proxy-based weighting model against direct measurements of cellular RNA content and independently known mixture compositions. Additional datasets and experimental mixtures could test whether the observed abundance-dependent errors and mismatch responses generalize across tissues, protocols, and target populations. Methodologically, future studies could incorporate sample-level observables of RNA composition, uncertainty-aware calibration, and models that jointly represent within-cluster yield distributions rather than relying primarily on cluster-level summaries. It would also be useful to evaluate marker-library-size dependence explicitly, compare calibration procedures with paired held-out uncertainty estimates, and design panels using a Pareto trade-off among panel size, detection, quantitative error, and yield-shift robustness. These directions would help determine when transcript-based assays can support reliable cell-abundance quantification and when they should be limited to detection or relative molecular measurement.

This work was generated by THE AI SCIENTIST (Lu et al., 2024).

REFERENCES

Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The AI Scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*, 2024.

Quantitative cell-fraction prediction — Target enriched high yield test time mismatch

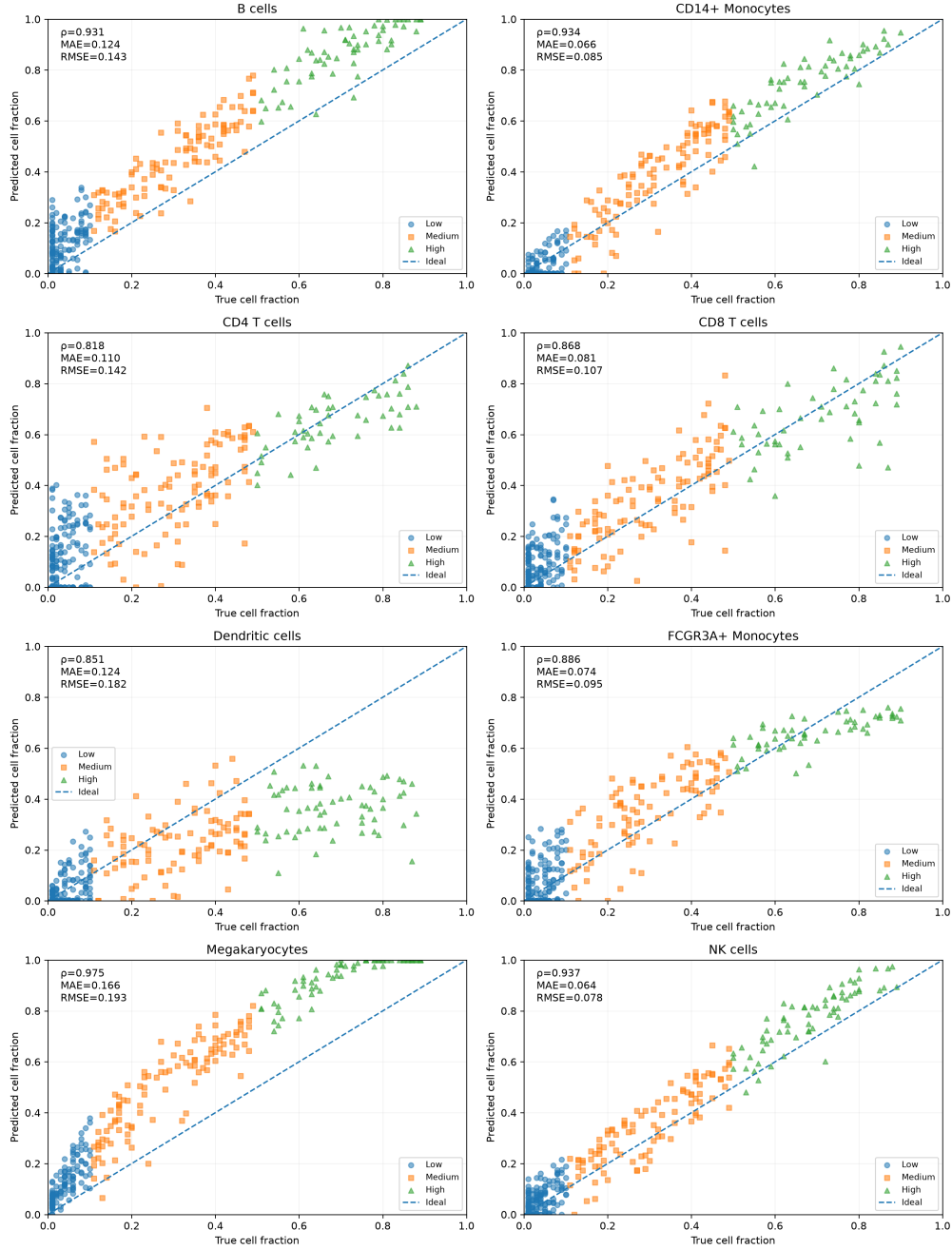


Figure 10: Predicted-versus-true target cell fractions under the prespecified $2.0\times$ target-enriched test-time RNA-yield condition.

Prediction residuals across abundance regimes — Target enriched high yield test time mismatch

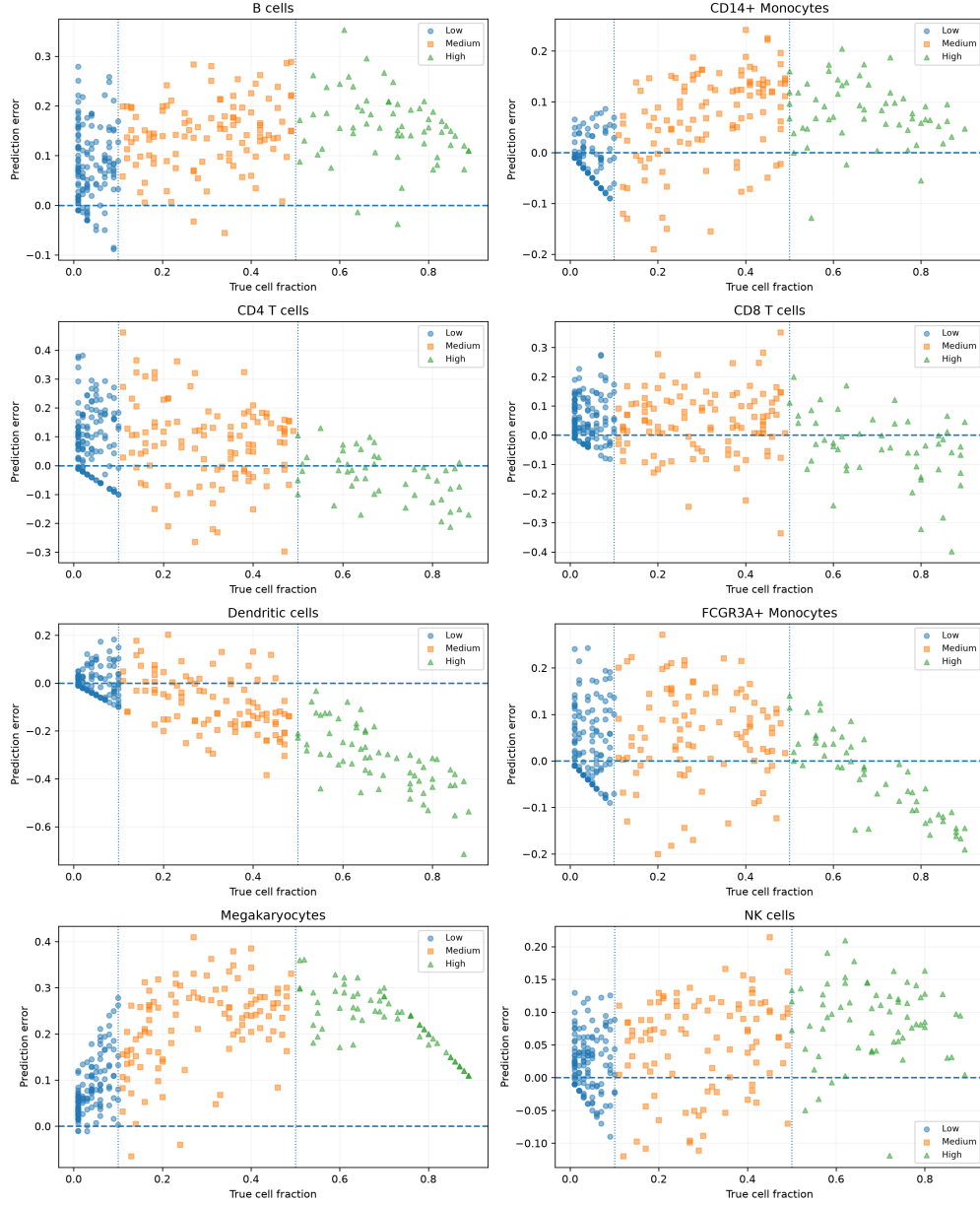


Figure 11: Cell-fraction residuals under the prespecified $2.0\times$ target-enriched test-time RNA-yield condition.

Quantitative cell-fraction prediction — Target enriched low yield test time mismatch

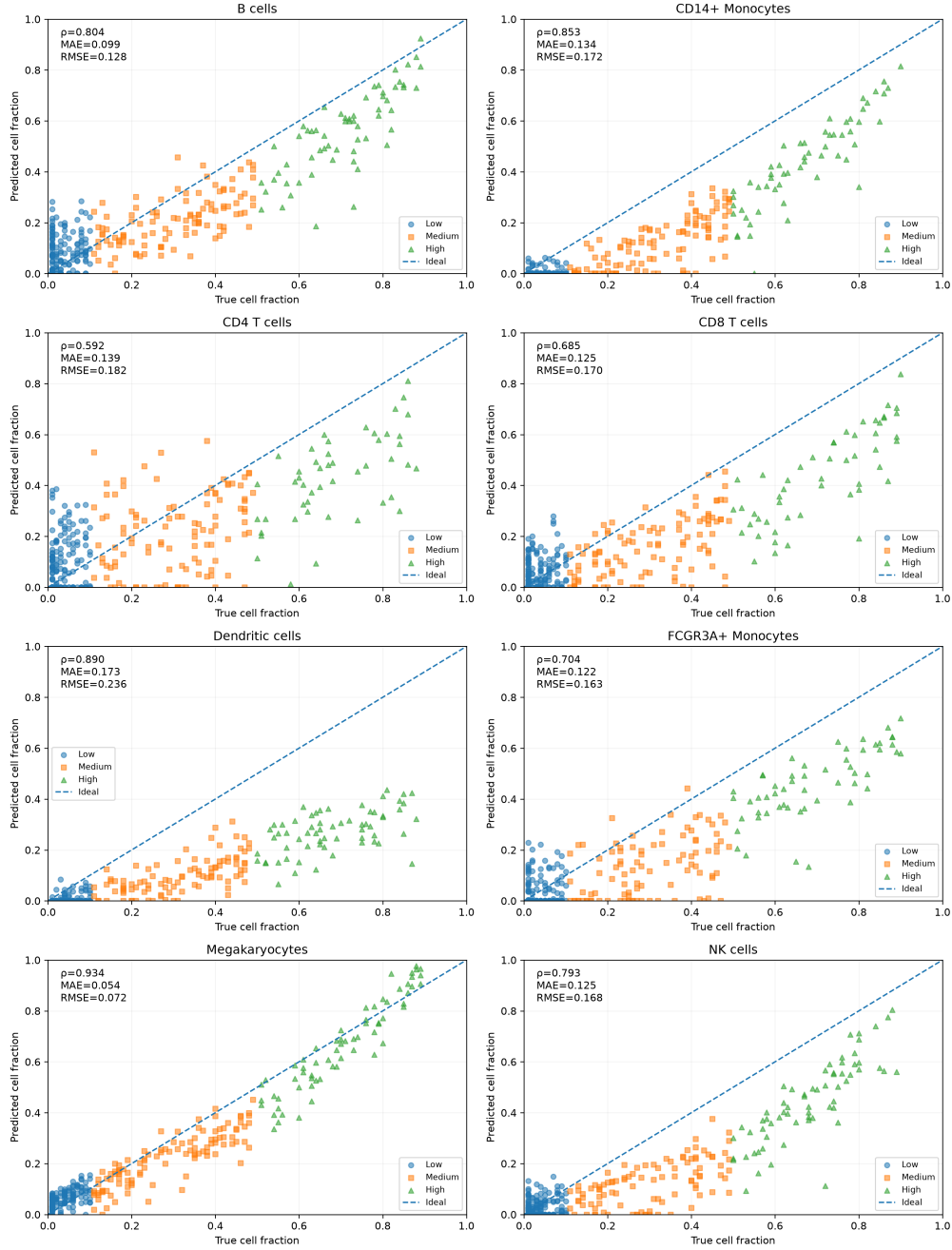


Figure 12: Held-out predicted versus true target cell fraction when target-cell RNA yield is reduced by a prespecified factor of 0.5.

Prediction residuals across abundance regimes — Target enriched low yield test time mismatch

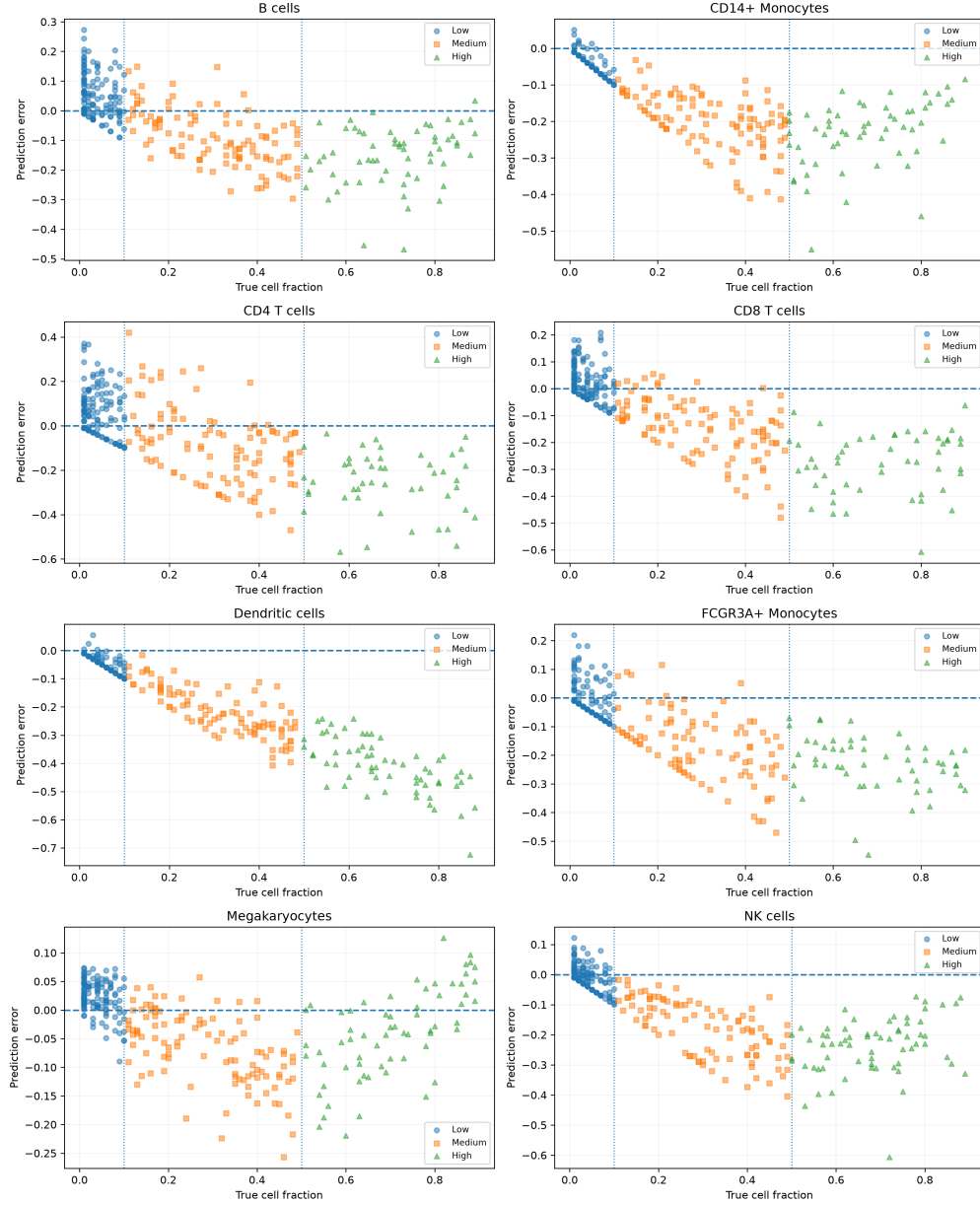


Figure 13: Cell-fraction residuals under the prespecified $0.5\times$ target-yield mismatch, with errors shown across abundance regimes.

Quantitative cell-fraction prediction — Equal cell/equal yield ablation

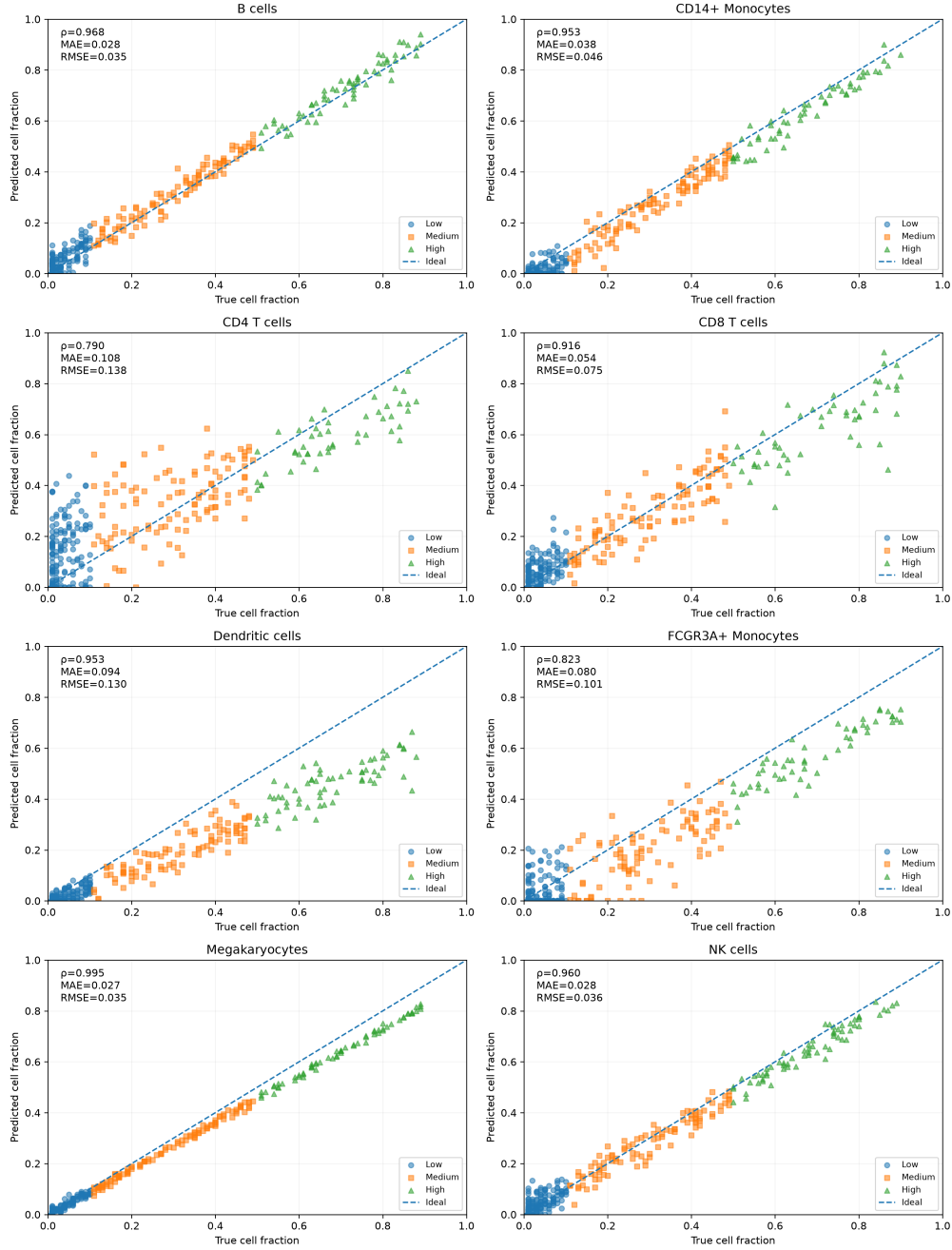


Figure 14: Held-out predicted versus true target cell fraction in the homogeneous equal-cell/equal-yield ablation.

Rare-target precision-recall performance — Experimental configuration

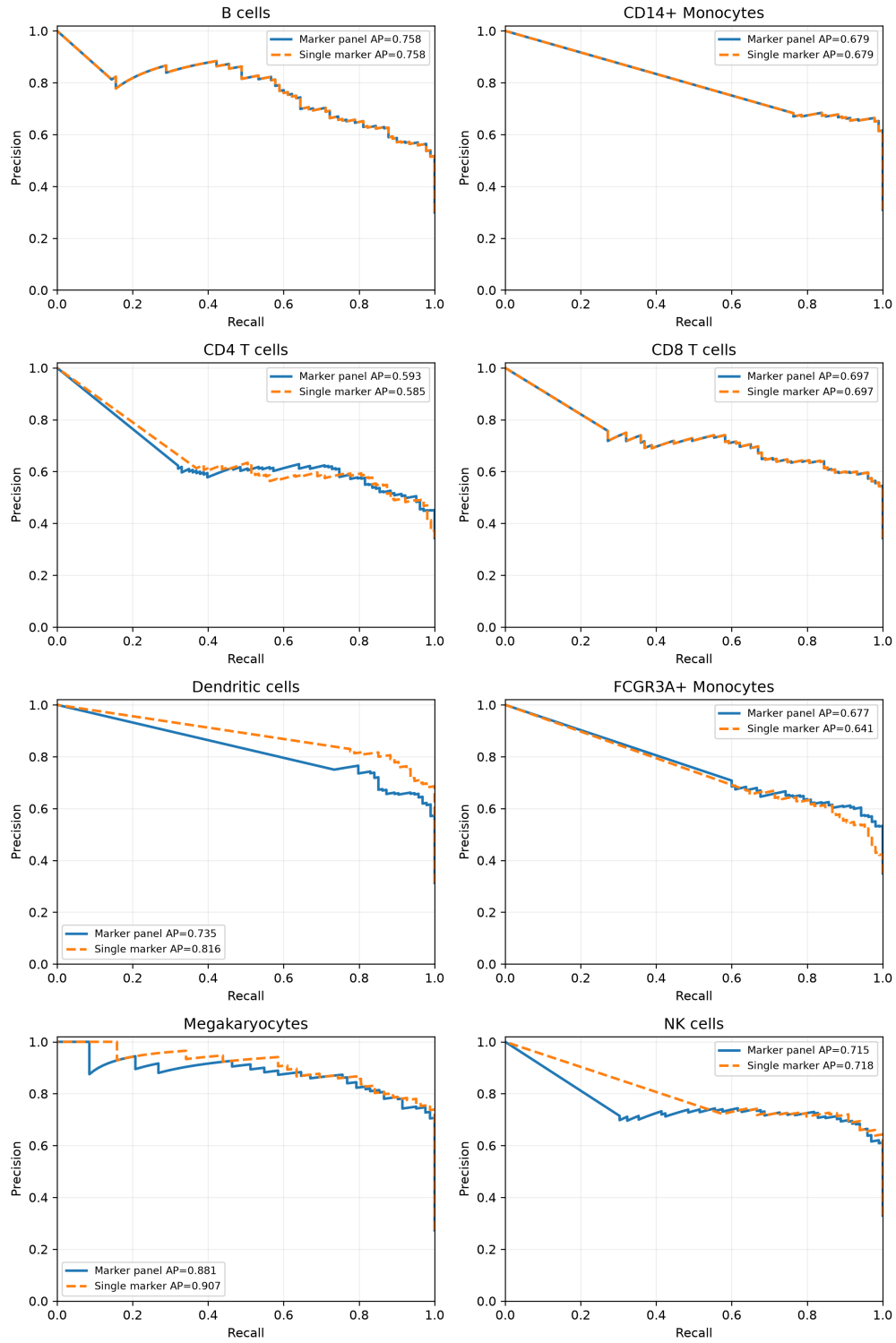
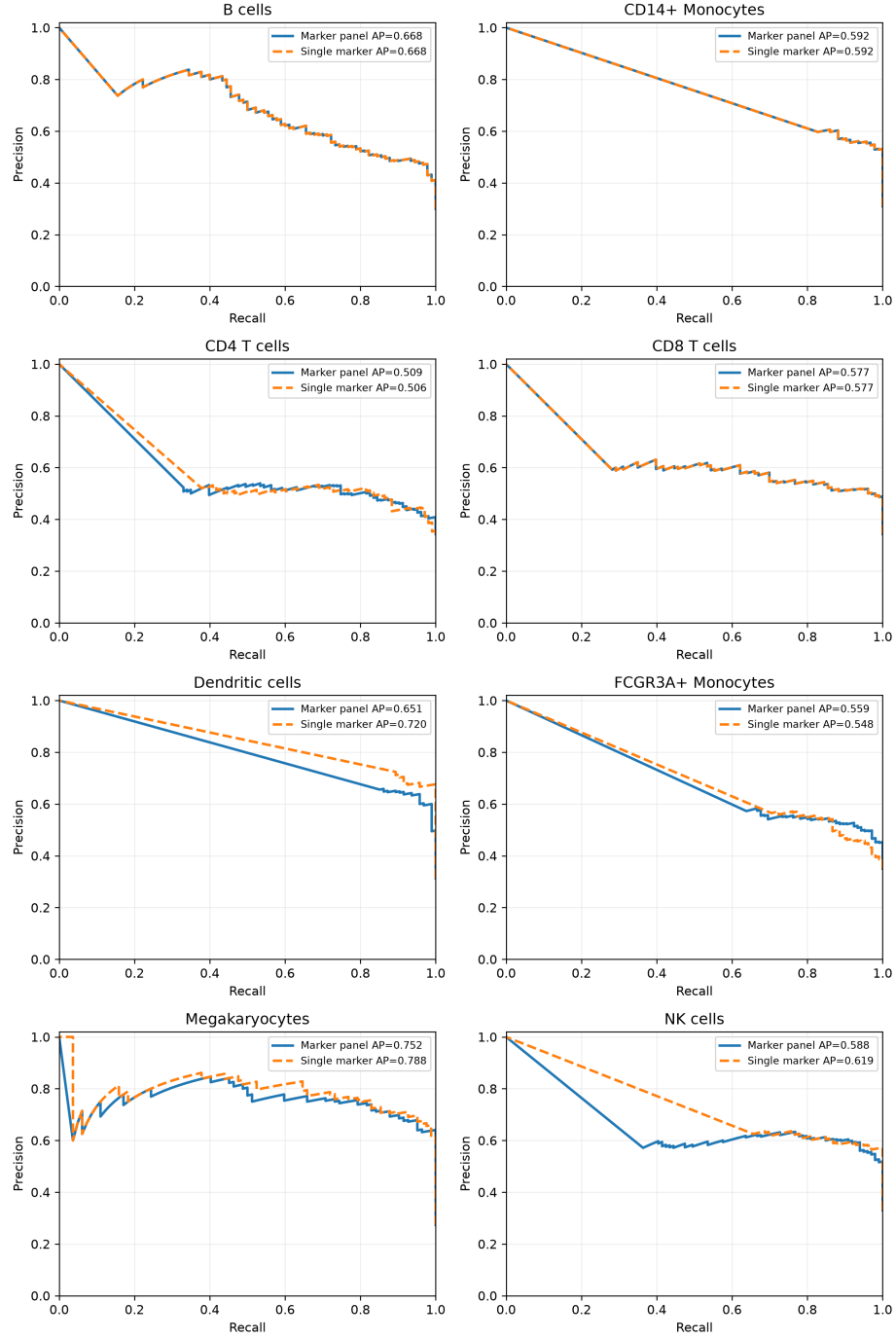


Figure 15: Per-cluster precision-recall curves for rare targets at or below 5% cell fraction under nominal heterogeneous RNA yield.

Rare-target precision-recall performance — Target enriched low yield test time mismatch

Figure 16: Per-cluster precision-recall curves for rare targets at or below 5% cell fraction under the prespecified $0.5\times$ target-yield mismatch.

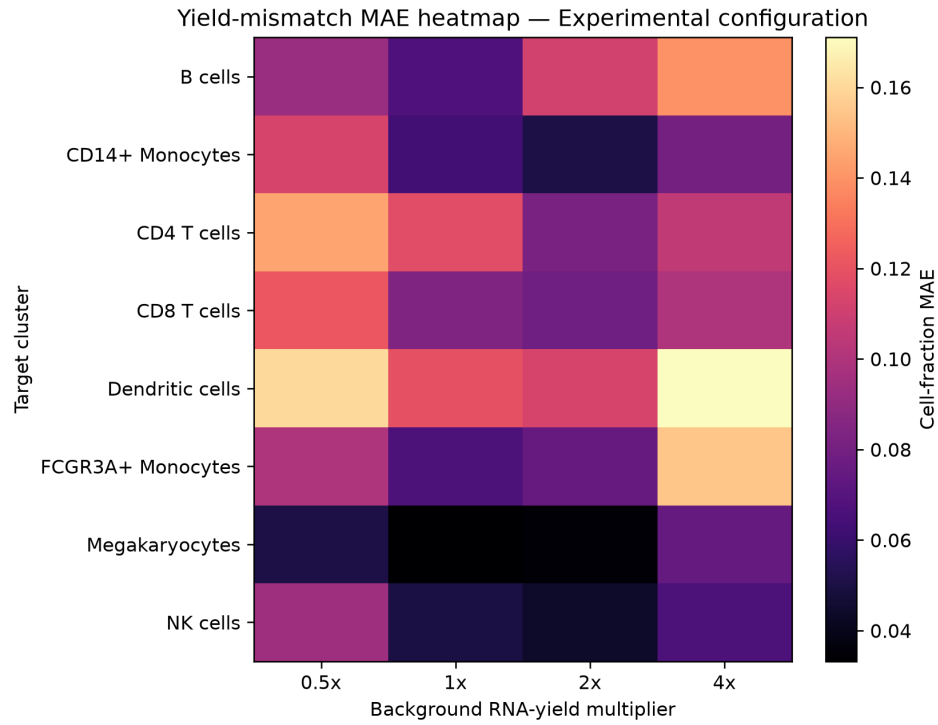


Figure 17: Cluster-by-yield-factor heatmap of held-out cell-fraction MAE under the recorded yield-mismatch stress test.

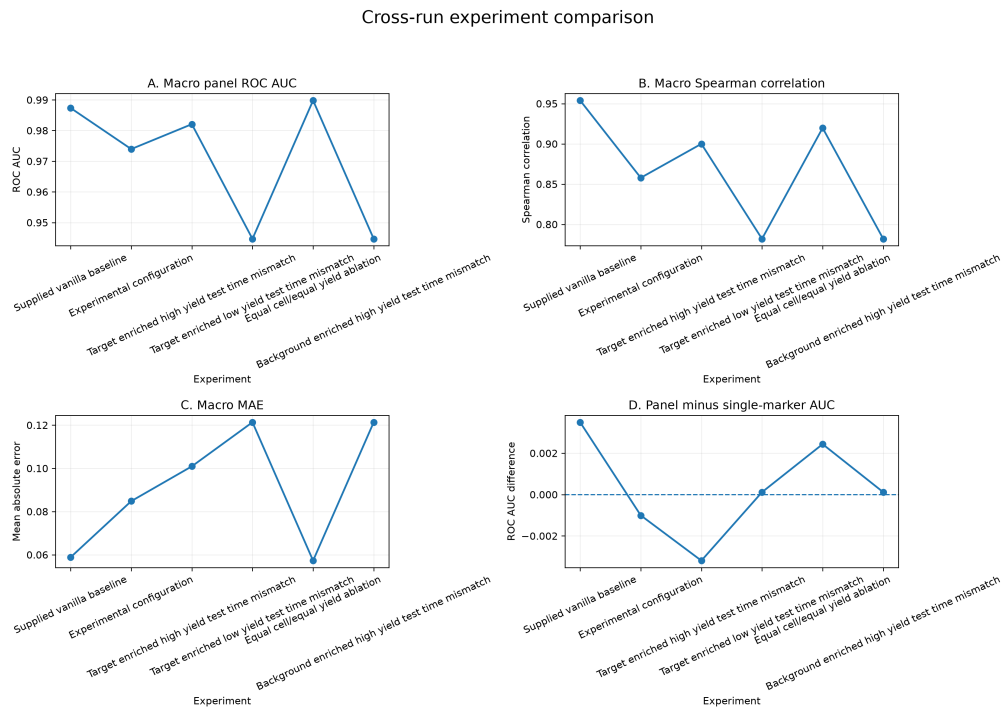


Figure 18: Comparison across the supplied baseline, nominal heterogeneous-yield, target-yield mismatch, and equal-yield regimes for macro panel ROC AUC, Spearman correlation, MAE, and panel-minus-single-marker AUC difference.